

Portfolio Risk Memorandum

Memo reference	72FB081EEB	Generated	2026-09-07 16:46 UTC
Population	112,717 loans scored	Basis	Loan-level, exposure-weighted

1. Request and retrieval

Question as asked: “What is our expected loss and concentration risk for California loans?”

Rows retrieved	113,008	Retrieval retries used	0
Loans scored	112,717	Result truncated	No

291 of 113,008 retrieved loans (0.26%) were not scored. The PD and LGD models require a complete feature row and these loans are missing at least one input: debt-to-income ratio missing or not positive (291). They are excluded rather than scored on an imputed value, so exposure, expected loss and regulatory capital in this memo are totals over the scored population. Concentration and the interest-rate view use all retrieved rows, since neither depends on a model score.

SQL executed against the portfolio database:

```
SELECT * FROM Loans JOIN Borrowers USING(loan_id) WHERE Borrowers.addr_state = 'CA'
```

2. Executive summary

Total exposure (EAD)	\$1,202,765,651	Expected loss	\$237,616,512
Exposure-weighted PD	21.34%	Exposure-weighted LGD	92.58%
Expected loss rate	19.76%	Average risk weight	197.50%
Total RWA	\$2,375,483,747	Capital reserve (8% of RWA)	\$190,038,700

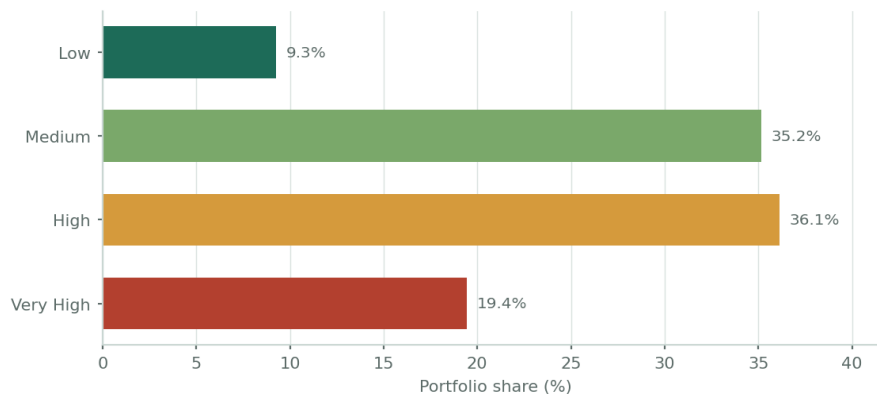
One or more limits are breached - see section 7.

3. Credit risk

Expected loss is computed per loan as $EL = PD \times LGD \times EAD$ and aggregated exposure-weighted, so a large exposure carries proportionally more weight than a small one. PD comes from a calibrated behavioural model scoring loans already on the book; LGD from a regression trained on realised recoveries, with loss measured against exposure at default rather than original loan size.

Metric	Value	Origin
Exposure-weighted PD	21.34%	pd_model_calibrated.joblib
Exposure-weighted LGD	92.58%	lgd_model.joblib
Total exposure (EAD)	\$1,202,765,651	Outstanding principal, portfolio database
Total expected loss	\$237,616,512	$EL = PD \times LGD \times EAD$
Expected loss rate	19.76%	Total EL / total EAD

Risk tier distribution

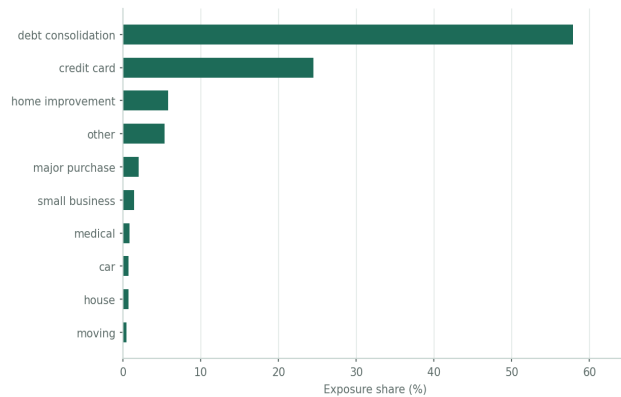


4. Concentration risk

Concentration is measured as the Herfindahl-Hirschman Index over exposure shares, reported on the conventional 0-10000 scale. Above 2500 is treated as highly concentrated, following the US DOJ merger-guideline convention.

Dimension	HHI (0-10000)	Assessment
Sector (loan purpose)	4,020	Highly Concentrated
Region (borrower state)	10,000	Highly Concentrated

By purpose



By region



5. Interest rate risk

Repricing gap per bucket is rate-sensitive assets minus rate-sensitive liabilities. Assets are bucketed by months remaining on each fixed-rate term loan, which is when a fixed loan effectively reprices. Buckets are measured from the portfolio's own latest issue date (2018-12-01), not the date this memo was generated, so a historical book is not reported as fully matured.

Net interest income (annual)	\$147,218,160	Net interest margin	12.20%
Earnings at risk (12m)	-\$14,425,579	Loan-to-deposit ratio	0.9524
Portfolio yield (observed)	12.87%	Deposit rate (assumed)	0.64%

The book earns a weighted-average contractual rate of **12.87%** on \$1,207,089,186 of loans, which is \$155,375,367 of annual interest income. Depositors are paid 5.00% of that rate - **0.64%** - on a deposit book sized at 1.05x the loans, costing \$8,157,207. Net interest income is the difference: \$147,218,160, or 12.20% of rate-sensitive assets.

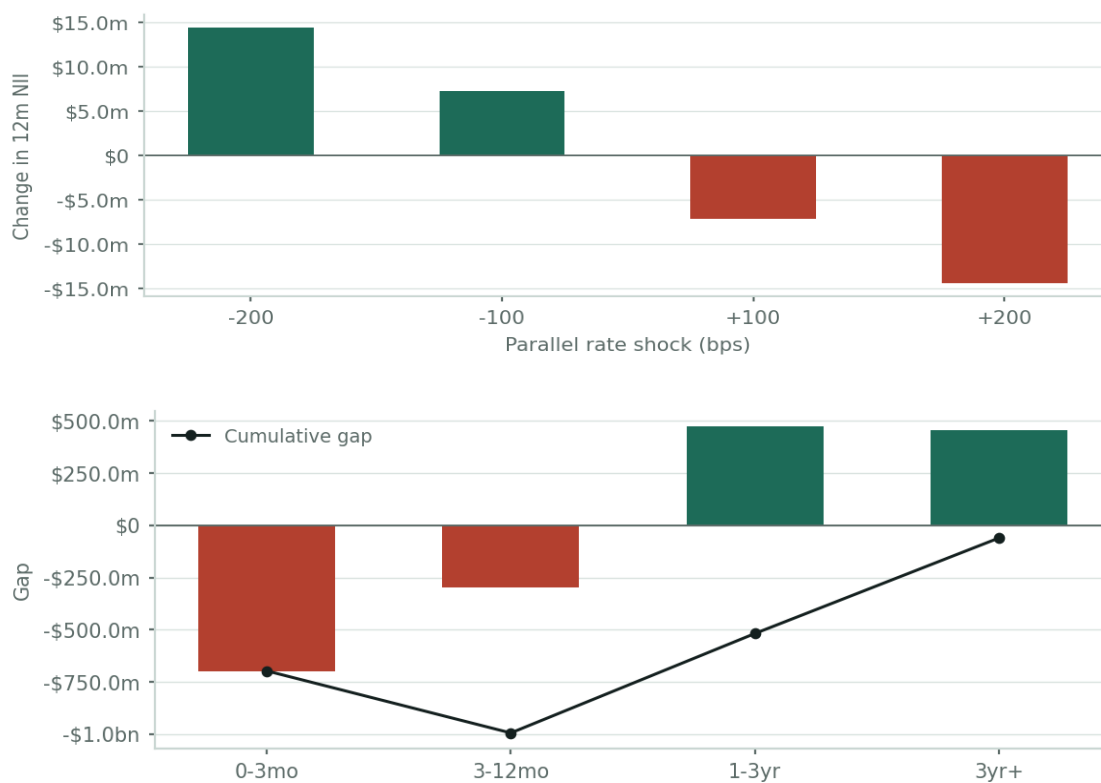
The book is **liability-sensitive** inside 12 months: more liabilities than assets reprice in that window, so rising rates compress net interest income before the loan book catches up.

Bucket	Rate-sensitive assets	Rate-sensitive liabilities	Gap	Cumulative gap
0-3mo	\$217,281	\$697,094,005	-\$696,876,724	-\$696,876,724
3-12mo	\$19,496,005	\$316,860,911	-\$297,364,907	-\$994,241,630
1-3yr	\$666,342,781	\$190,116,547	\$476,226,234	-\$518,015,396
3yr+	\$521,033,119	\$63,372,182	\$457,660,937	-\$60,354,459
Total	\$1,207,089,186	\$1,267,443,645	-\$60,354,459	

Earnings at risk under parallel shifts in the rate curve. Each bucket's gap is exposed to the shock only for the part of the next 12 months that follows its repricing, so the near buckets are weighted by their midpoint (0-3mo at 0.875 of the year, 3-12mo at 0.375) and anything repricing beyond 12 months contributes nothing. The headline earnings-at-risk figure is the worst of these.

Parallel shock	Change in 12m NII	12m NII after	Change (%)
-200 bps	\$14,425,579	\$161,643,739	+9.80%
-100 bps	\$7,212,790	\$154,430,950	+4.90%
+100 bps	-\$7,212,790	\$140,005,370	-4.90%
+200 bps	-\$14,425,579	\$132,792,580	-9.80%

Documented assumption. This portfolio has no liability data - it is a lending book, not a full bank balance sheet. The asset side is real throughout: balances, contractual rates, terms and issue dates all come from the retrieved rows. The deposit side is modelled: a book sized at 1.05x the loans, spread across the four buckets at 55% / 25% / 15% / 5% to reflect a retail profile weighted toward short-term repricing, and paid 5.00% of the rate earned on the loans. That pass-through gives a deposit rate near the sub-1% US retail savings rates actually observed over this portfolio's 2013-2018 origination window. The gap table, the earnings figures and the loan-to-deposit limit check all inherit that assumption and should be read as illustrative rather than as this bank's real funding position.



6. Regulatory capital (Basel III IRB)

Quantity	Value
Exposure at default (EAD)	\$1,202,765,651
Exposure-weighted correlation (R)	0.0361
Exposure-weighted capital requirement (K)	0.1580
Average risk weight	197.50%
Risk-weighted assets (RWA)	\$2,375,483,747
Capital reserve at 8% of RWA	\$190,038,700
Exposures included	112,717

Formula as implemented, verified against the primary BIS source:

$$R = 0.03 \times (1 - e^{(-35 \times PD)}) / (1 - e^{-35}) + 0.16 \times [1 - (1 - e^{(-35 \times PD)}) / (1 - e^{-35})]$$

$$K = LGD \times N[(G(PD) + \sqrt{R / (1 - R)} \times G(0.999)) / \sqrt{1 - R}] - PD \times LGD$$

$$RWA = K \times 12.5 \times EAD$$

Basel III CRE31.16, IRB risk-weight function for the **Other Retail** exposure class - the correct class for unsecured personal instalment loans (not mortgages, not qualifying revolving). Retail exposures carry no maturity adjustment: maturity is subsumed in the correlation assumption, per the BIS QIS3 FAQ. Input floors of 0.05% on PD and 30% on LGD are applied per CRE32.58 before the formula is evaluated. Where K evaluates negative it is clipped to zero, meaning the exposure attracts no regulatory capital.

7. Compliance findings

Limit	Actual	Threshold	Test	Source	Status
Expected loss rate	0.1976	0.05	Ceiling	Internal	BREACHED
Concentration HHI (0-10000)	10,000	2,500	Ceiling	Internal	BREACHED

Limit	Actual	Threshold	Test	Source	Status
Loan-to-deposit ratio	0.9524	1.1	Ceiling	Internal	Within limit
PD floor - other retail	0.2134	0.0005	Floor	Basel III	Within limit
LGD floor - unsecured other retail	0.9258	0.3	Floor	Basel III	Within limit

Attribution for each finding:

Expected loss rate - Internal risk policy threshold (not itself a Basel limit). Elevated PD/LGD estimates directly increase risk-weighted assets and regulatory capital requirements under Basel III CRE31.16 -- see regulatory_capital_citation below.

Concentration HHI (0-10000) - Internal risk policy threshold -- not derived from Basel III. Basel's credit risk framework (CRE30-32) does not set portfolio-level sector concentration limits; concentration risk is addressed under each bank's own risk appetite framework, and separately, for single-counterparty exposures, under Basel's Large Exposures framework (not applicable to sector-level retail concentration).

Limit checking is fully deterministic - no model judgement is involved at this step. Basel-sourced floors carry the exact source paragraph they were taken from; internal risk-appetite limits are labelled as internal policy rather than being given a regulatory citation they do not have.

8. Source attribution

Every figure in this memo traces to exactly one of four origins. Nothing below is an unattributed estimate.

Figure	Class of source	Attribution
Loan and borrower attributes	Portfolio data	SQLite portfolio database, Loans joined to Borrowers; 113,008 rows returned by the SQL in section 1
Exposure at default (EAD)	Portfolio data	Each loan's current outstanding principal
Probability of default (PD)	Trained model	pd_model_calibrated.joblib - XGBoost with isotonic calibration, leakage-audited, behavioural (scores loans already on the book, not new applicants)
Loss given default (LGD)	Trained model	lgd_model.joblib - XGBoost regression on realised recoveries, loss measured against exposure at default
Expected loss	Deterministic formula	$EL = PD \times LGD \times EAD$, aggregated exposure-weighted
Concentration (HHI)	Deterministic formula	Sum of squared exposure shares on the 0-10000 scale; 2500 threshold per US DOJ merger-guideline convention
Repricing gap - asset side	Portfolio data	Months remaining on each fixed-rate term loan, measured from the portfolio's own latest issue date
Portfolio yield, interest income	Portfolio data	Weighted-average contractual int_rate on the retrieved loans - observed, not assumed
Repricing gap - liability side	Documented assumption	Synthetic retail deposit profile (55/25/15/5) sized at 1.05x the loan book - no liability data exists in this portfolio
Deposit rate, interest expense, NII, NIM	Documented assumption	Depositors paid 5% of the observed asset yield; every figure downstream of that inherits the assumption
Earnings at risk (12m)	Deterministic formula	Worst change in 12m net interest income across +/-100 and +/-200bp parallel shocks, gaps weighted by midpoint exposure within the year
Loan-to-deposit ratio	Internal policy	Loans / assumed deposits, checked against an internal 1.1 ceiling - Basel addresses funding stability through the NSFR, not an LTD cap
Correlation R, capital requirement K, RWA	Regulatory source	Basel III CRE31.16, IRB risk-weight function for Other Retail exposures
PD and LGD input floors	Regulatory source	Basel III CRE32.58 - PD floor 0.05%, LGD floor 30% for unsecured other retail
Expected loss rate and HHI thresholds	Internal policy	Risk_Limits table - illustrative internal risk-appetite defaults, not regulatory requirements

Scope note. The underlying portfolio is US unsecured consumer instalment loans originated between October 2013 and December 2018. The PD model is behavioural and supports portfolio-level reporting, not individual accept or reject decisions. This memo is generated automatically from a single query and is not a substitute for a reviewed credit opinion.